Principal Components, Score Matrices, and Orthogonal Projection

Spectral Stability of Equity Covariance

Mathematical Foundations of Data Science
University of Vienna

November 2026

Abstract

We derive the basic constructions of principal component analysis for an asset-return data matrix. Beginning with centering and sample covariance, we identify variance along a direction as a covariance quadratic form and obtain the first principal direction through constrained maximization. We then express the maximizing direction in an eigenbasis and construct principal-component scores, orthogonal projectors, and reconstructed observations for a leading k-dimensional subspace.

1 Centering an asset-return matrix

Let $r_1, \dots, r_n \in \mathbb{R}^d$ denote observed return vectors for d assets over $n \geq 2$ trading days. Arrange the uncentered observations as rows of $R \in \mathbb{R}^{n\times d}$, and define

$$\bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i, \qquad X = R - \mathbf{1}_n \bar{r}^{\mathsf{T}} \in \mathbb{R}^{n\times d}, \tag{1}$$

where $\mathbf{1}_n = (1, \dots, 1)^{\mathsf{T}} \in \mathbb{R}^n$. The ith row of X is x_i^{T}, with $x_i = r_i - \bar{r}$. Equivalently,

$$X = H_n R, \qquad H_n = I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}. \tag{2}$$

Indeed, since $\mathbf{1}_n^{\mathsf{T}}\mathbf{1}_n = n$,

$$\mathbf{1}_n^{\mathsf{T}} X = \mathbf{1}_n^{\mathsf{T}} R - \frac{1}{n}(\mathbf{1}_n^{\mathsf{T}}\mathbf{1}_n)\mathbf{1}_n^{\mathsf{T}} R = 0,$$

so the sample mean of the centered observations is zero.

Definition 1.1 (Empirical covariance)**.** We use the convention

$$\widehat{\Sigma} = \frac{1}{n} X^{\mathsf{T}} X = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^{\mathsf{T}} \in \mathbb{R}^{d\times d}. \tag{3}$$

If the original r_i are independent and identically distributed with finite second moments, the estimator $X^{\mathsf{T}}X/(n-1)$ is unbiased for the population covariance. Mean-centering imposes the constraint $\sum_i x_i = 0$, leaving $n-1$ degrees of freedom in the usual sample-covariance calculation. The $1/n$ and $1/(n-1)$ conventions differ by a positive scalar: their eigenvectors coincide and their eigenvalues differ by the factor $n/(n-1)$.

2 Directional variance and the first principal component

Fix a direction $u \in \mathbb{R}^d$. The scalar return of a projected observation is

$$z_i = u^{\mathsf{T}} r_i, \qquad \bar{z} = u^{\mathsf{T}}\bar{r}, \tag{4}$$

and its empirical variance is

$$\frac{1}{n}\sum_{i=1}^{n}(z_i-\bar{z})^2=\frac{1}{n}\sum_{i=1}^{n}\big(u^{\mathsf{T}}(r_i-\bar{r})\big)^2$$
$$=u^{\mathsf{T}}\left(\frac{1}{n}\sum_{i=1}^{n}x_i x_i^{\mathsf{T}}\right)u=\boxed{u^{\mathsf{T}}\widehat{\Sigma}u}. \tag{5}$$

The same identity follows from $\|Xu\|^2/n = u^{\mathsf{T}}X^{\mathsf{T}}Xu/n$.

Without a normalization constraint, the maximization of $u^{\mathsf{T}}\widehat{\Sigma}u$ is unbounded whenever there is a direction of positive variance: replacing u with cu multiplies the objective by c^2. Thus we restrict attention to unit vectors.

Proposition 2.1 (Variance-maximizing direction)**.** *Let $\widehat{\Sigma}$ be symmetric and positive semidefinite. The solution of*

$$\max_{u\in\mathbb{R}^d,\,\|u\|=1} u^{\mathsf{T}}\widehat{\Sigma}u \tag{6}$$

is any unit eigenvector associated with the largest eigenvalue λ_1 of $\widehat{\Sigma}$. The maximal variance equals λ_1.

Proof. First write the Lagrangian

$$L(u,\lambda)=u^{\mathsf{T}}\widehat{\Sigma}u-\lambda(u^{\mathsf{T}}u-1).$$

Because $\widehat{\Sigma}$ is symmetric, stationarity gives

$$\nabla_u L=2\widehat{\Sigma}u-2\lambda u=0 \quad\Longrightarrow\quad \widehat{\Sigma}u=\lambda u.$$

Multiplying by u^{T} and using $u^{\mathsf{T}}u=1$ shows that the stationary objective value is its corresponding eigenvalue λ.

To identify the maximum, use an orthonormal eigenbasis $v_1,\dots,v_d$ of $\widehat{\Sigma}$, with $\lambda_1\ge\cdots\ge\lambda_d\ge 0$. Every unit vector can be written

$$u=\sum_{j=1}^{d}a_j v_j,\qquad \sum_{j=1}^{d}a_j^2=1.$$

Orthogonality and $\widehat{\Sigma}v_j=\lambda_j v_j$ yield

$$u^{\mathsf{T}}\widehat{\Sigma}u=\sum_{j=1}^{d}\lambda_j a_j^2\le\lambda_1\sum_{j=1}^{d}a_j^2=\lambda_1. \tag{7}$$

Taking $u=v_1$ attains the bound. □

Remark 2.2. If λ_1 has multiplicity greater than one, the maximizing direction is not unique; every unit vector in the leading eigenspace is a maximizer.

3 A two-asset variance example

Consider the covariance matrix

$$\widehat{\Sigma}=\begin{pmatrix}0.04 & 0.018\\ 0.018 & 0.01\end{pmatrix}. \tag{8}$$

For the unit directions

$$u_1=\begin{pmatrix}1\\0\end{pmatrix},\quad u_2=\begin{pmatrix}0\\1\end{pmatrix},\quad u_3=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},\quad u_4=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix},$$

direct multiplication gives

$$\begin{aligned} &u_1^{\mathsf{T}}\widehat{\Sigma}u_1=0.04, && u_2^{\mathsf{T}}\widehat{\Sigma}u_2=0.01,\\ &u_3^{\mathsf{T}}\widehat{\Sigma}u_3=\tfrac{1}{2}(0.058+0.028)=0.043, && u_4^{\mathsf{T}}\widehat{\Sigma}u_4=\tfrac{1}{2}(0.022-0.008)=0.007. \end{aligned} \tag{9}$$

The 45° direction u_3 therefore retains more variance than either coordinate direction. These four choices are illustrative candidate directions; the exact maximizer is given by the leading eigenvector from Proposition 2.1.

4 Scores, projections, and reconstruction

Let $v_1, \dots, v_k$ be orthonormal principal directions, with $1 \le k \le d$, and put

$$V_k = [v_1, \dots, v_k] \in \mathbb{R}^{d \times k}, \qquad V_k^{\mathsf{T}} V_k = I_k. \tag{10}$$

The *principal-component score vector* of observation x_i is $V_k^{\mathsf{T}} x_i \in \mathbb{R}^k$. Collecting these vectors as rows produces the score matrix

$$\boxed{Z = XV_k \in \mathbb{R}^{n \times k}}, \qquad Z_{ij} = x_i^{\mathsf{T}} v_j. \tag{11}$$

Proposition 4.1 (Projector onto the principal subspace)**.** *The matrix*

$$\boxed{P_k = V_k V_k^{\mathsf{T}} \in \mathbb{R}^{d \times d}} \tag{12}$$

is the orthogonal projector onto $\mathcal{U}_k = \operatorname{span}\{v_1, \dots, v_k\}$*, with*

$$P_k^{\mathsf{T}} = P_k, \qquad P_k^2 = P_k, \qquad \operatorname{im}(P_k) = \mathcal{U}_k, \qquad \ker(P_k) = \mathcal{U}_k^{\perp}.$$

Proof. The orthonormality of V_k gives

$$P_k^2 = V_k (V_k^{\mathsf{T}} V_k) V_k^{\mathsf{T}} = V_k V_k^{\mathsf{T}} = P_k, \qquad P_k^{\mathsf{T}} = P_k.$$

For any $x \in \mathbb{R}^d$, $P_k x = V_k (V_k^{\mathsf{T}} x)$ lies in the column space of V_k. Conversely, for every $x = V_k a \in \mathcal{U}_k$,

$$P_k x = V_k (V_k^{\mathsf{T}} V_k) a = V_k a = x.$$

Hence $\operatorname{im}(P_k) = \mathcal{U}_k$. Finally,

$$P_k x = 0 \quad \Longleftrightarrow \quad V_k^{\mathsf{T}} x = 0,$$

since V_k has orthonormal columns. The latter condition means that x is orthogonal to each v_j, so $\ker(P_k) = \mathcal{U}_k^{\perp}$. □

The projection of an observation is

$$\widehat{x}_i = P_k x_i = \sum_{j=1}^{k} \langle x_i, v_j \rangle v_j. \tag{13}$$

Applying this row by row yields the *reconstruction matrix*

$$\boxed{\widehat{X}_k = XP_k = XV_k V_k^{\mathsf{T}} = ZV_k^{\mathsf{T}}.} \tag{14}$$

This is the rank-at-most-k reconstruction obtained by projecting onto the leading principal subspace. Its optimality in squared reconstruction error is established by the corresponding retained-variance argument for the leading k-dimensional eigenspace.